

# PANKAJ KUMAR GAJPALLA

☎ +91-8815242412 | ✉ pankajsahu23102003@gmail.com | 🐙 Github | in LinkedIn | </> Leetcode

## Summary

---

Enthusiastic Computer Science student with strong interest in Machine Learning, Computer Vision, and AI-based problem solving. Passionate about building real-world projects using Python, OpenCV, and ML algorithms, and continuously improving problem-solving and analytical skills.

## Education

---

|                                                                                          |                |
|------------------------------------------------------------------------------------------|----------------|
| <b>Kalinga University, Raipur (C.G)</b><br><i>B.Tech in Computer Science Engineering</i> | 2022 – Present |
| <b>DPS, Rajnandgaon (C.G)</b><br><i>Higher Secondary (Class 12)</i>                      | 2021<br>89.0%  |

## Technical Skills

---

Languages: Python, C++, C

Machine Learning & AI: Supervised & unsupervised Learning, Classification, Regression, Clustering

Model Evaluation & Feature Engineering

Computer Vision (OpenCV): Image Processing, Face Detection & Recognition, Emotion Detection, Object Detection (Contours, Haar Cascades)

Libraries & Tools: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, OpenCV, TensorFlow / Keras (basic)

Other Tools: Git & GitHub, Streamlit, Kaggle

Soft Skills: Problem Solving, Communication, Team Collaboration

## Projects

---

### Real-Time Facial Emotion Detection (OpenCV + ML)

- Developed a real-time facial emotion detection system using OpenCV and Python.
- Used Haar Cascade for face detection and trained ML model for emotion classification.
- Detected emotions such as Happy, Sad, Angry, Neutral from live webcam feed.

### Object Detection System Using OpenCV

- Built an object detection system using contour detection and image thresholding.
- Applied image preprocessing techniques like grayscale conversion, edge detection, and dilation.
- Successfully detected and labeled objects in real-time video streams.

### Machine Learning Prediction Web App (Streamlit)

- Created an end-to-end ML project including EDA, feature engineering, model training, and evaluation.
- Deployed the trained model using Streamlit for real-time predictions.
- Implemented model serialization using Pickle.

## Internships / Training

---

Completed hands-on training in Machine Learning and Data Analytics.

Worked on real datasets from Kaggle for model building and evaluation.

## Achievements & Activities

---

Participated in Hackathons and Coding Competitions.

Strong interest in competitive programming and problem solving.

State-Level chess player